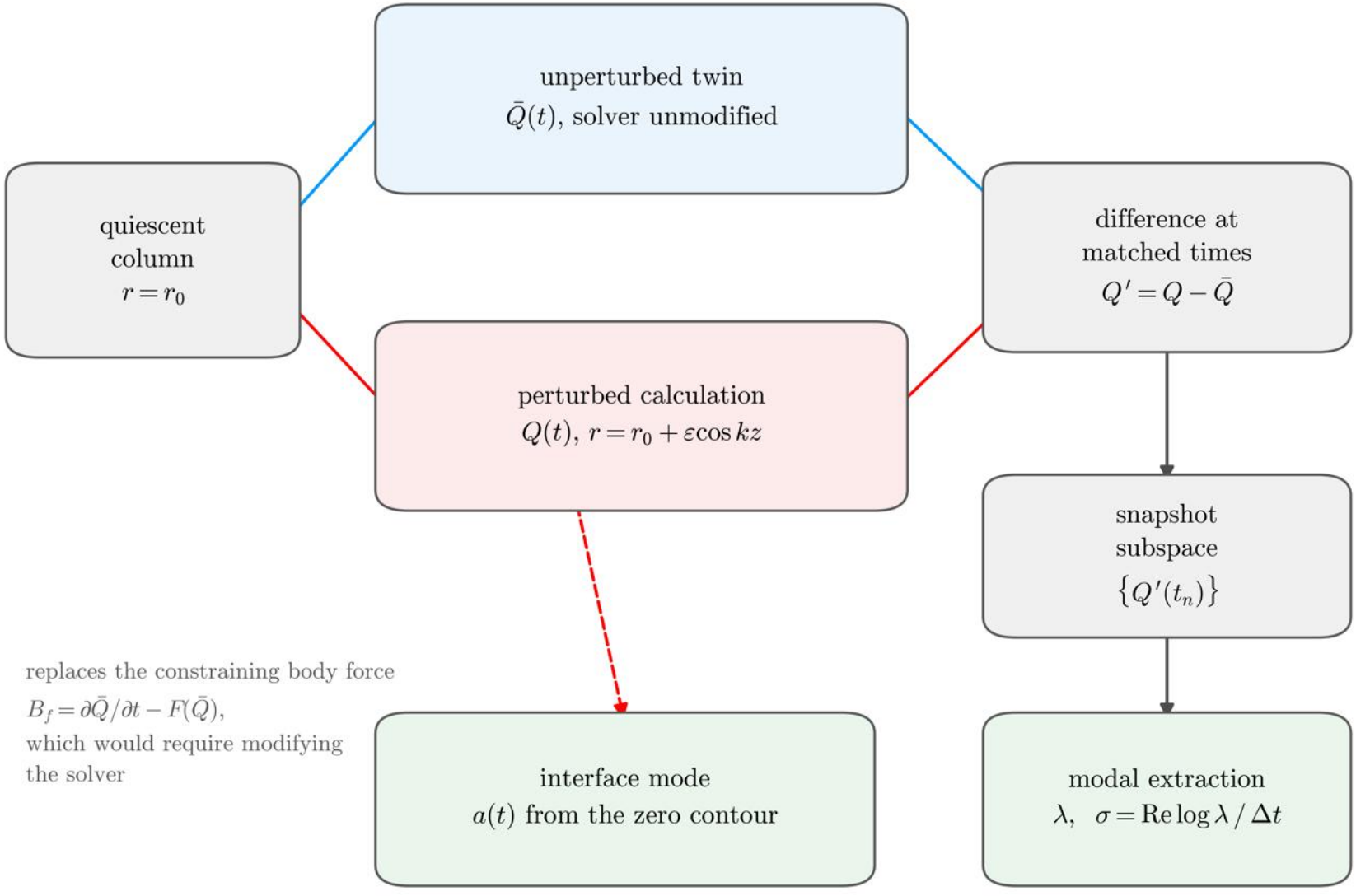
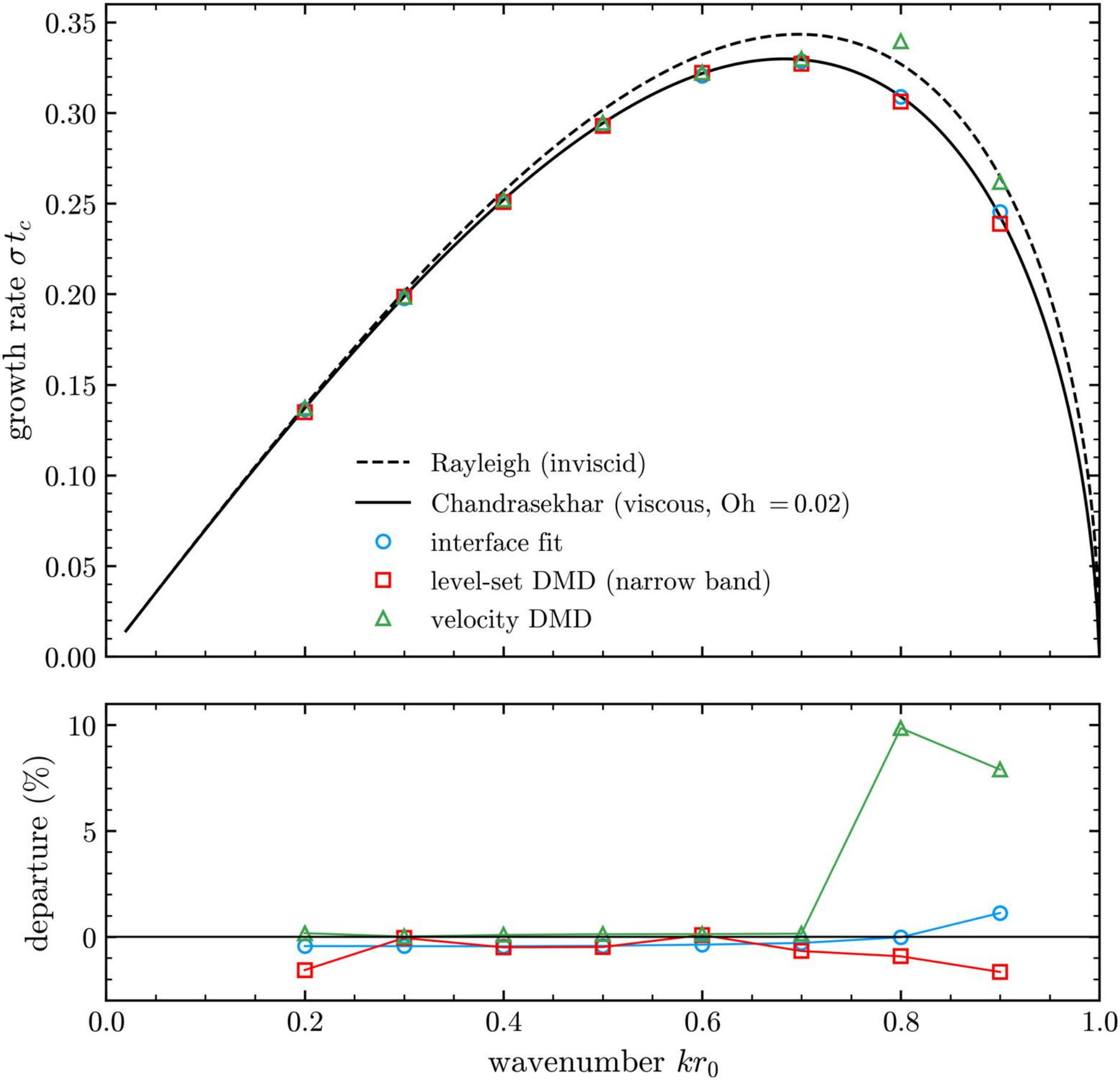
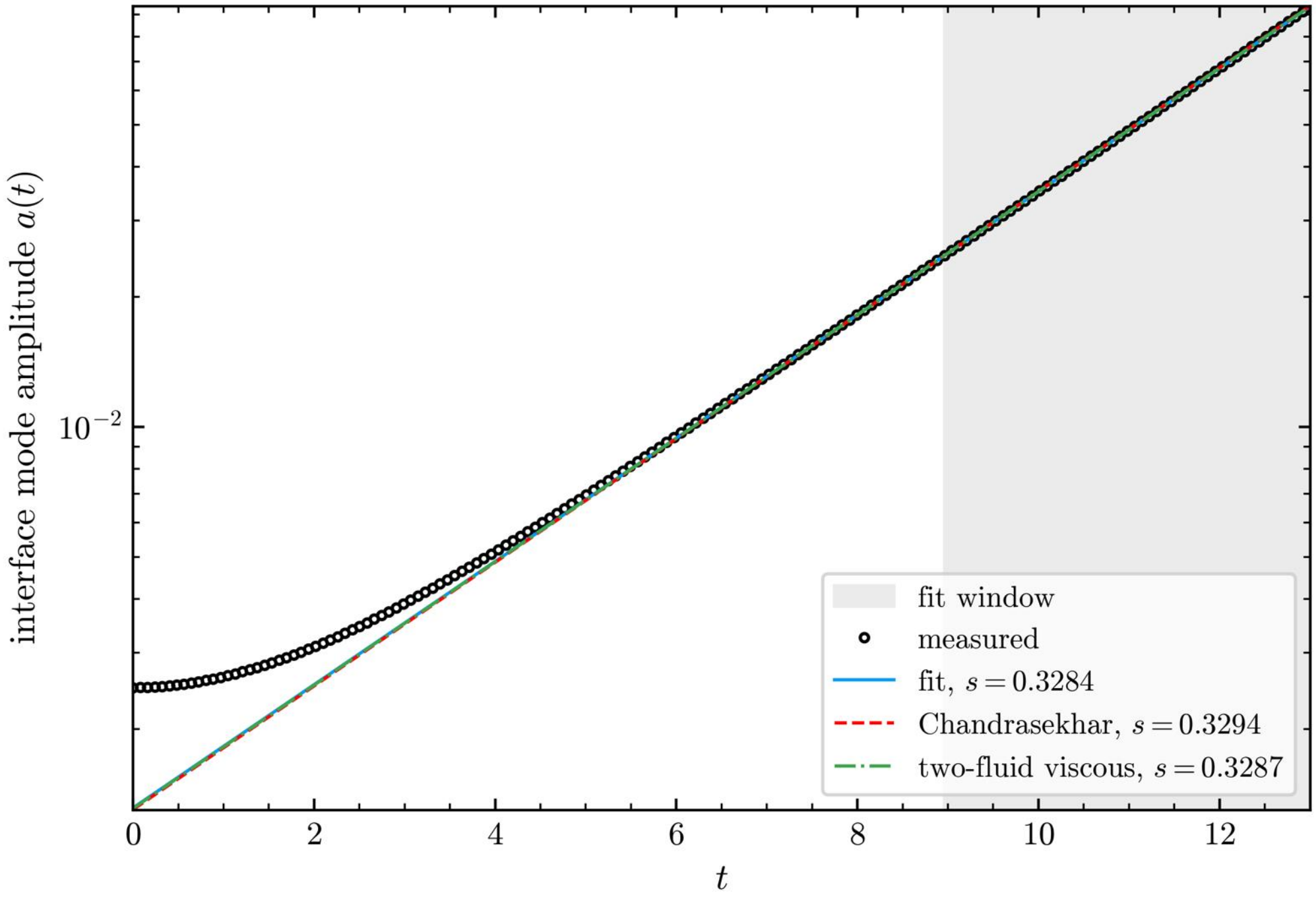
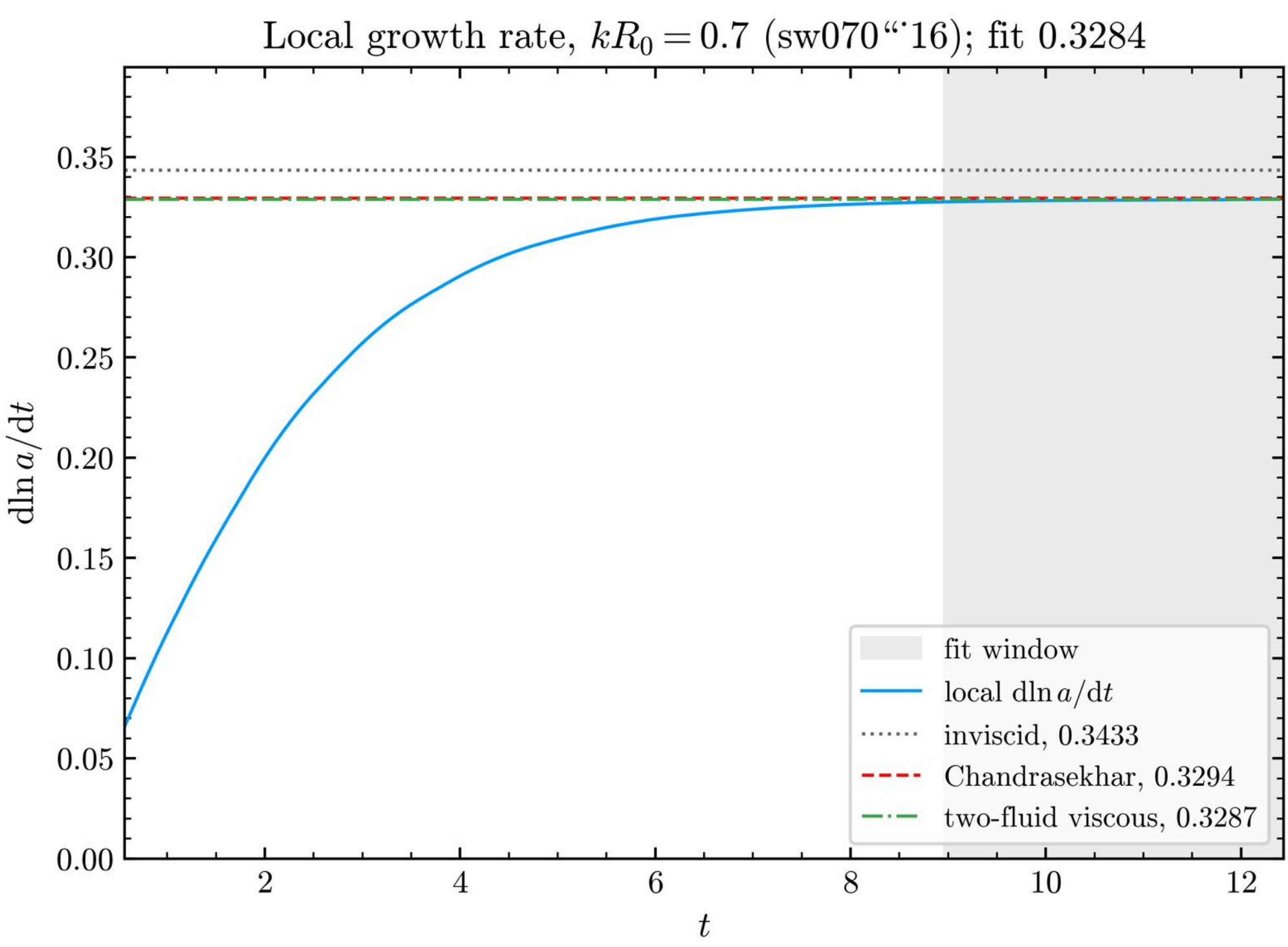
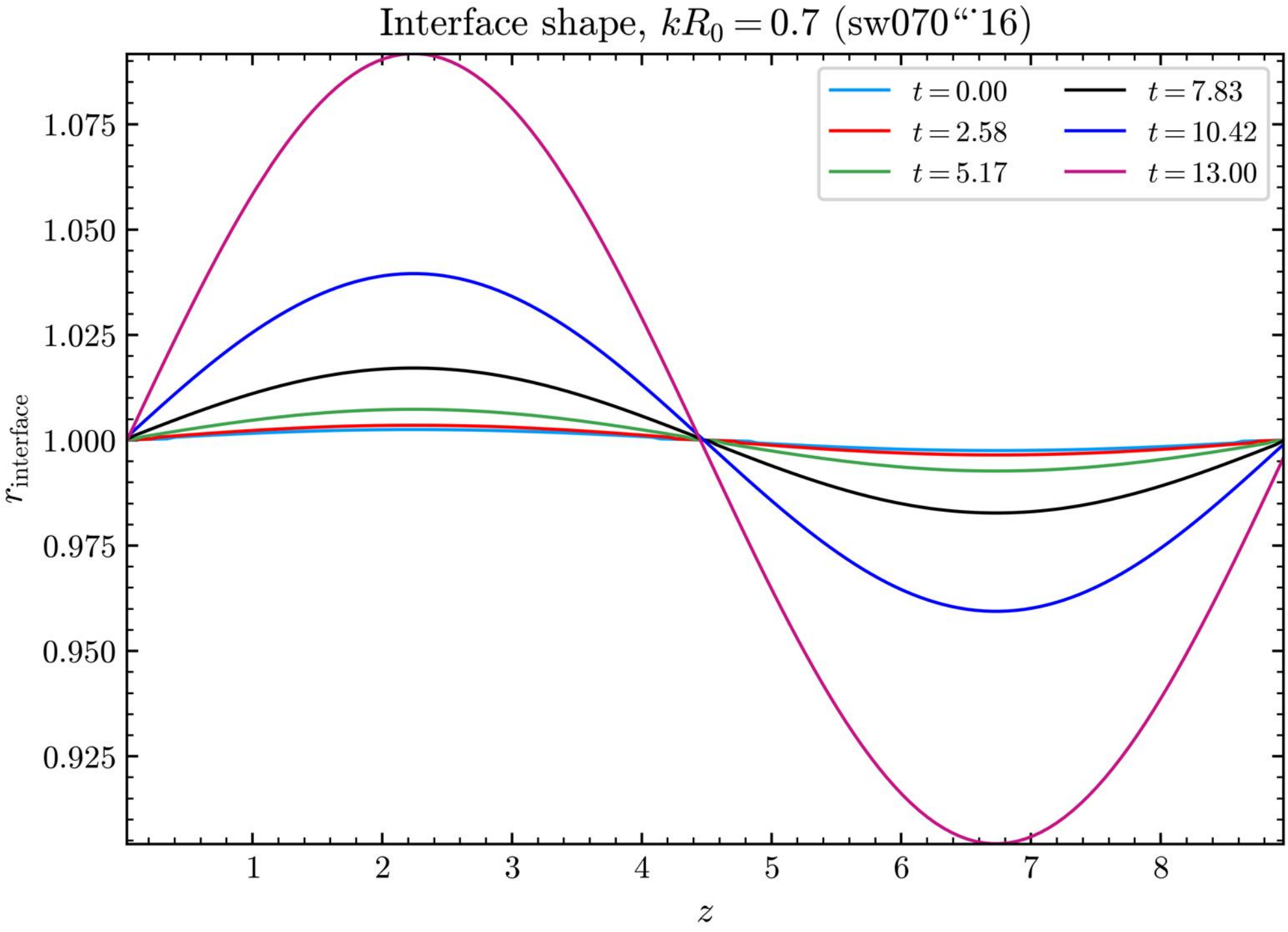
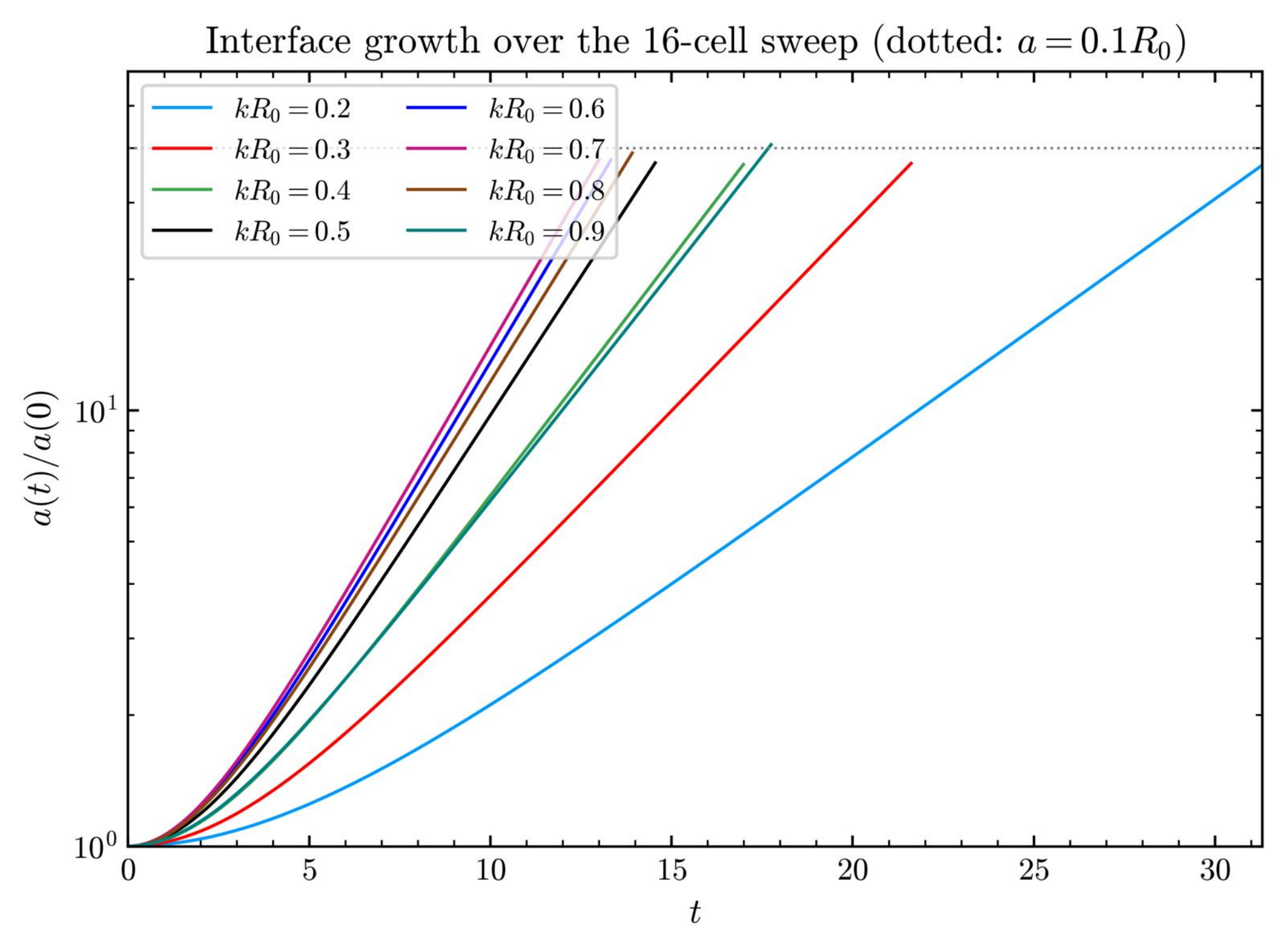

unperturbed twin
$\bar{Q}(t)$, solver unmodified
quiescent column
$r = r_0$
difference at matched times
$Q' = Q - \bar{Q}$
perturbed calculation
$Q(t)$, $r = r_0 + \varepsilon \cos kz$
snapshot subspace
$\{Q'(t_n)\}$
replaces the constraining body force
$B_f = \partial \bar{Q}/\partial t - F(\bar{Q})$,
which would require modifying the solver
interface mode
$a(t)$ from the zero contour
modal extraction
λ, $\sigma = \mathrm{Re} \log \lambda / \Delta t$
the interface observable needs no twin,
which halves the cost of a growth-rate study

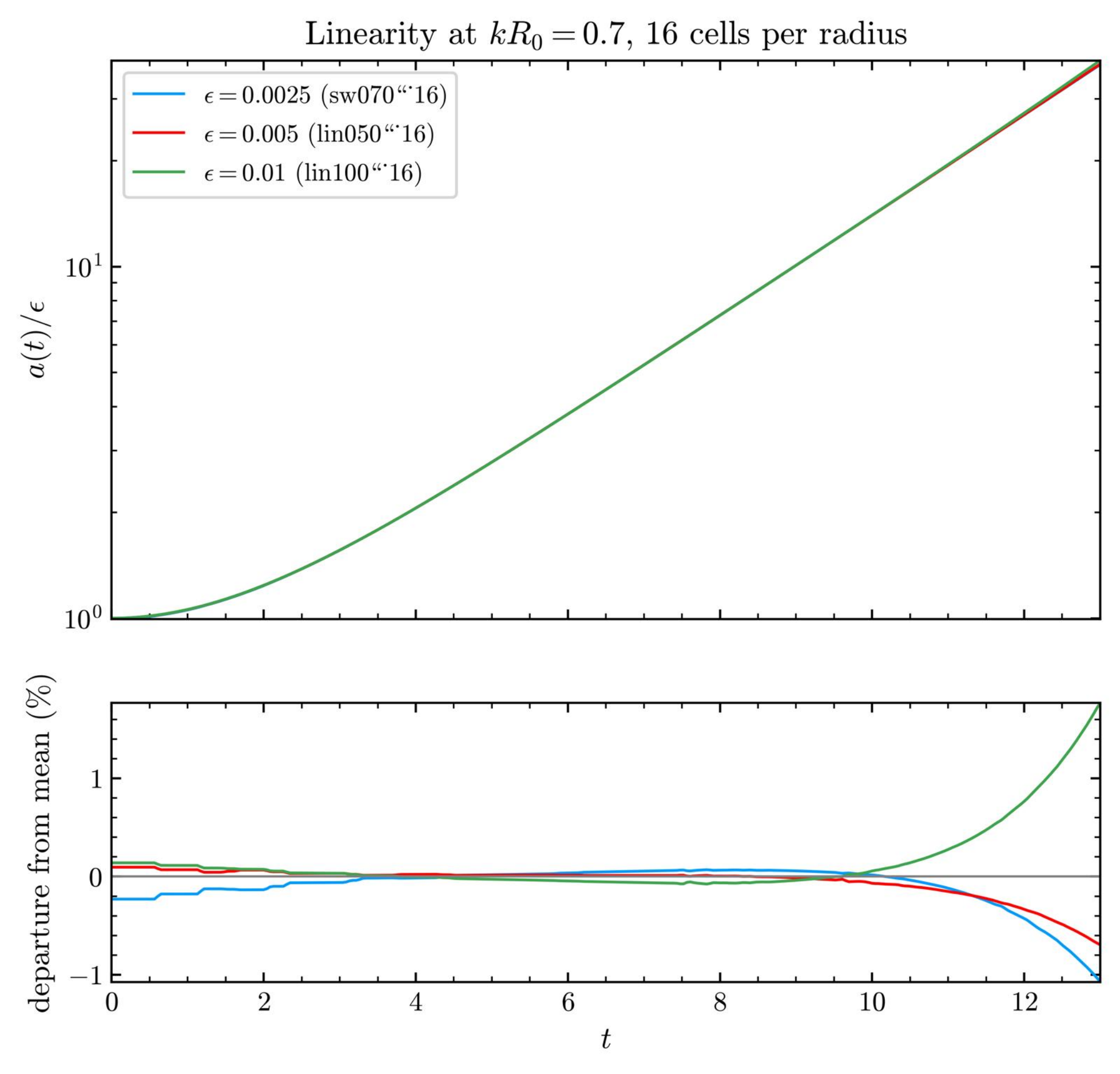

growth rate σt_c
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Rayleigh (inviscid)
Chandrasekhar (viscous, Oh $= 0.02$)
interface fit
level-set DMD (narrow band)
velocity DMD
departure (%)
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
wavenumber kr_0

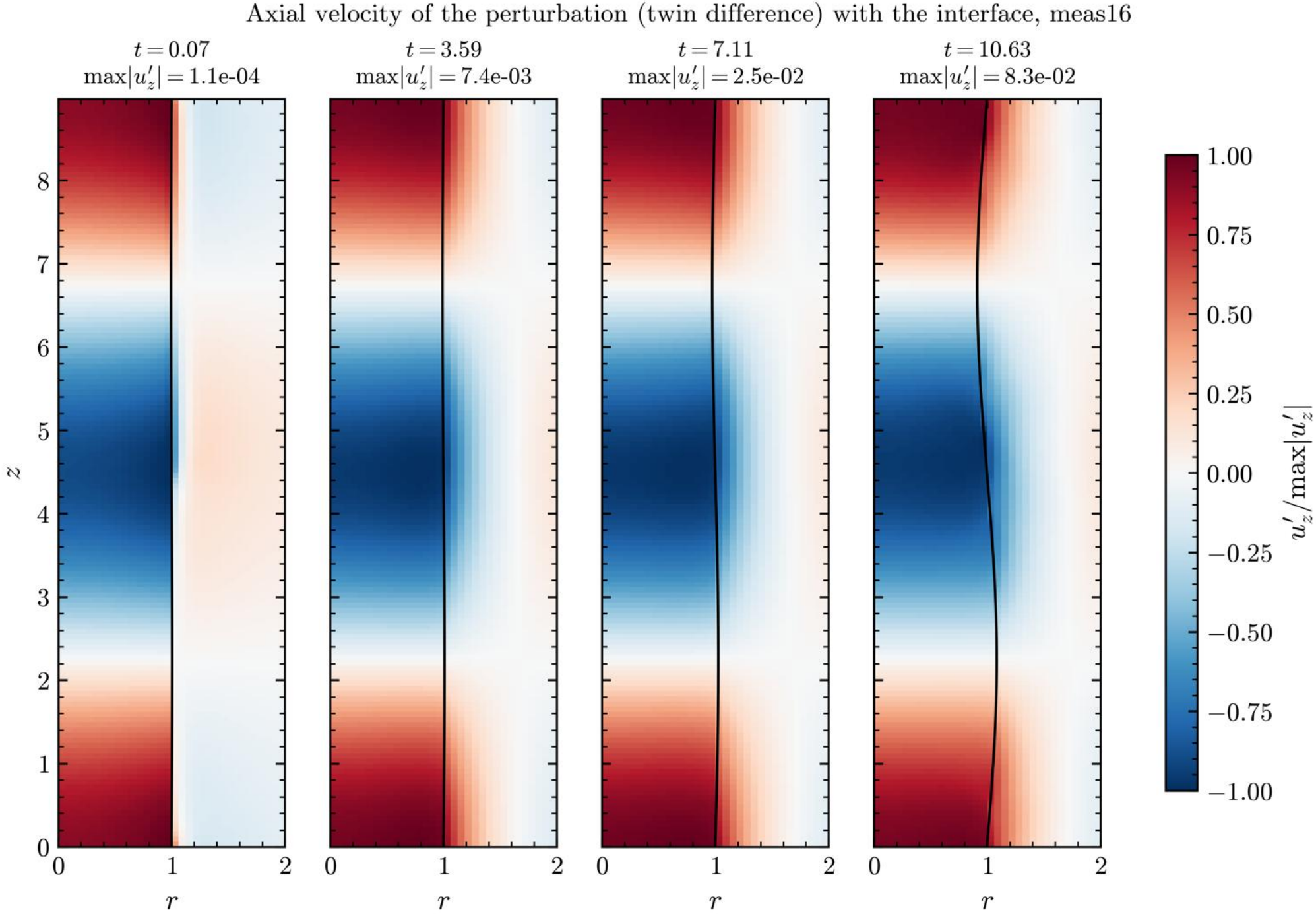

Interface growth, $kR_0 = 0.7$ (sw070‘‘16)
interface mode amplitude $a(t)$
10^{-2}
0
2
4
6
8
10
12
t
fit window
measured
fit, $s = 0.3284$
Chandrasekhar, $s = 0.3294$
two-fluid viscous, $s = 0.3287$

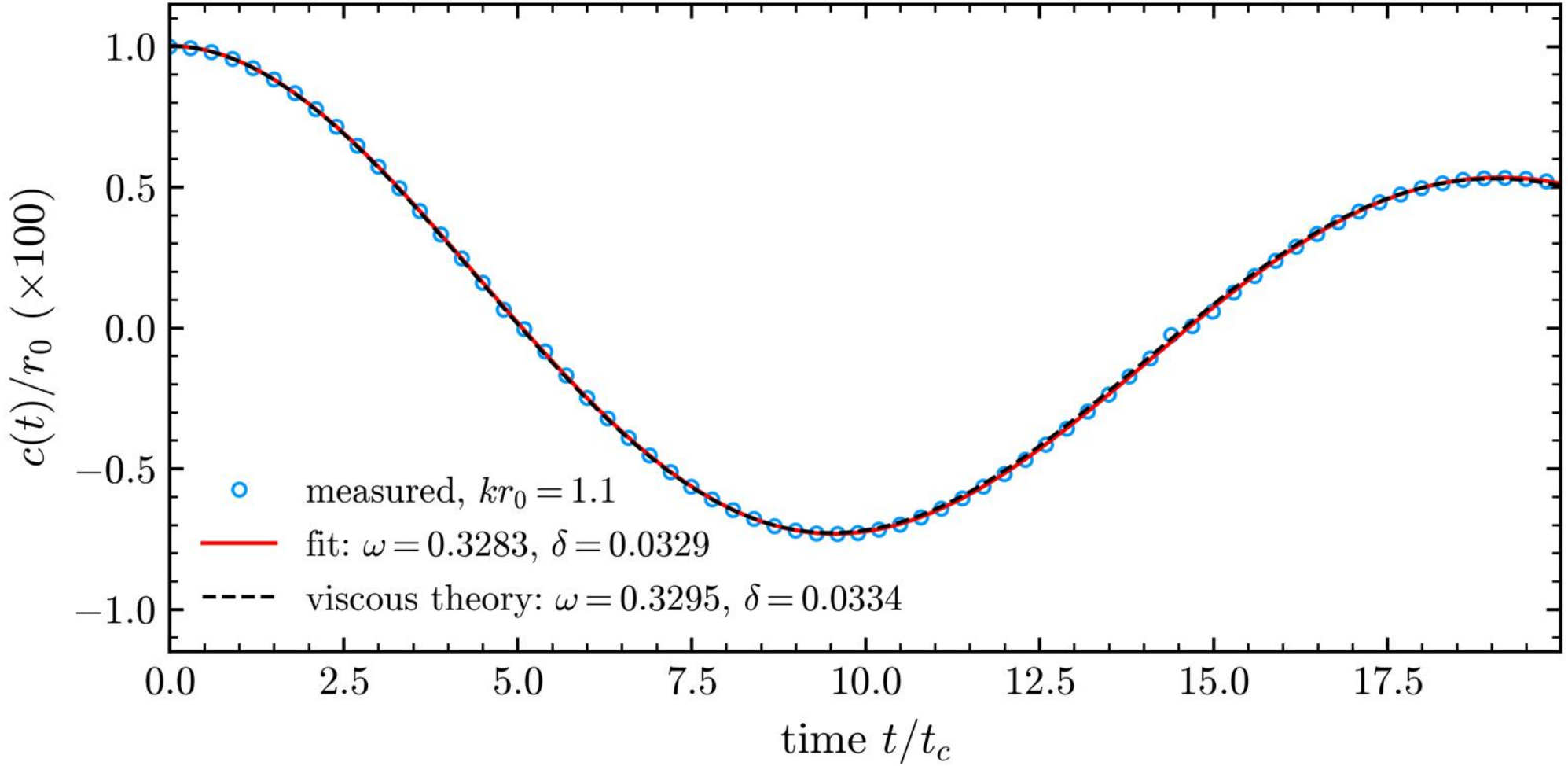

Local growth rate, $kR_0 = 0.7$ (sw070‘16); fit 0.3284
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$\mathrm{d}\ln a/\mathrm{d}t$
2
4
6
8
10
12
t
fit window
local $\mathrm{d}\ln a/\mathrm{d}t$
inviscid, 0.3433
Chandrasekhar, 0.3294
two-fluid viscous, 0.3287

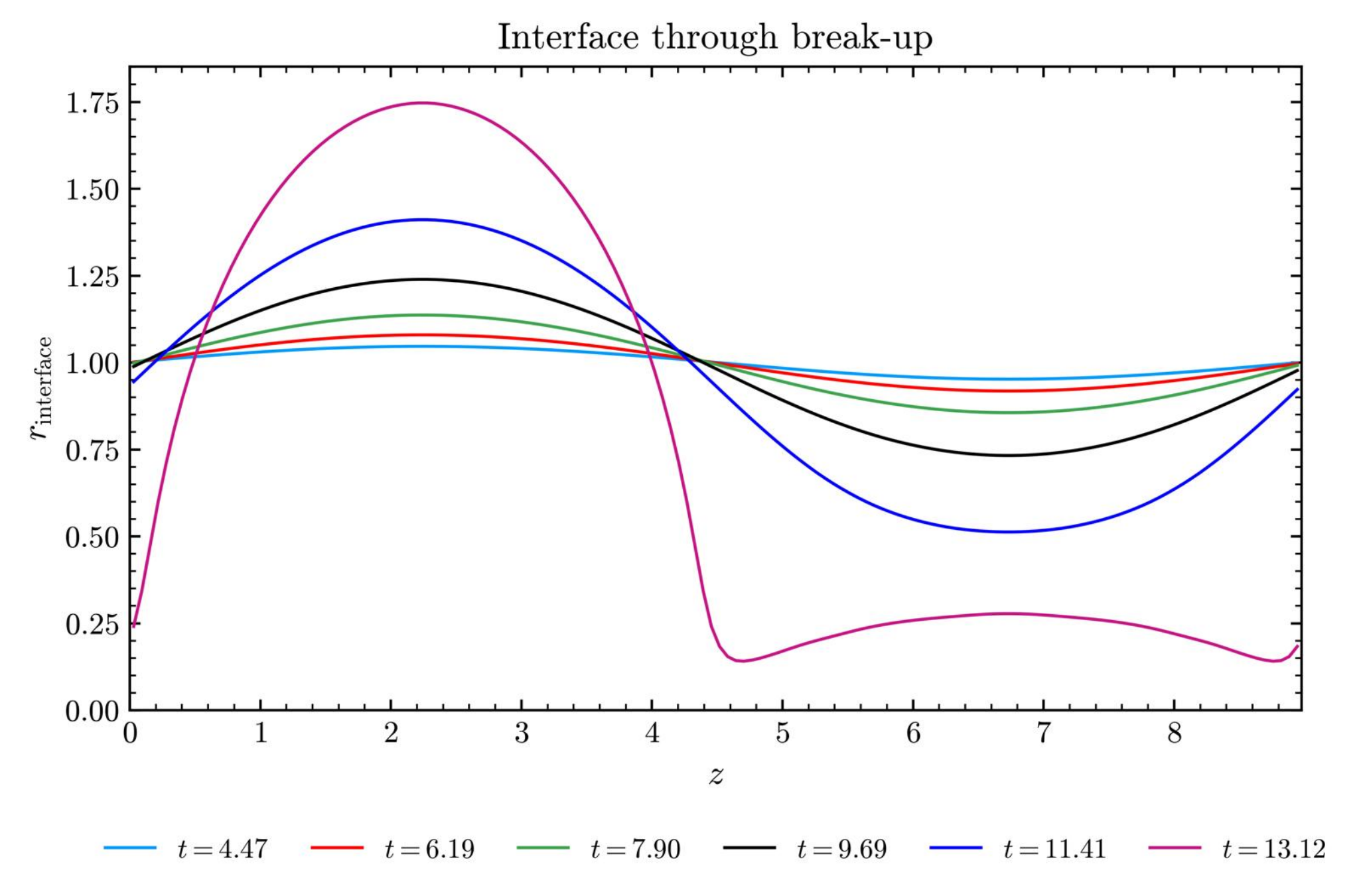

Interface shape, $kR_0 = 0.7$ (sw070‘‘16)
$t = 0.00$
$t = 2.58$
$t = 5.17$
$t = 7.83$
$t = 10.42$
$t = 13.00$
1.075
1.050
1.025
1.000
0.975
0.950
0.925
$r_{\text{interface}}$
1
2
3
4
5
6
7
8
z

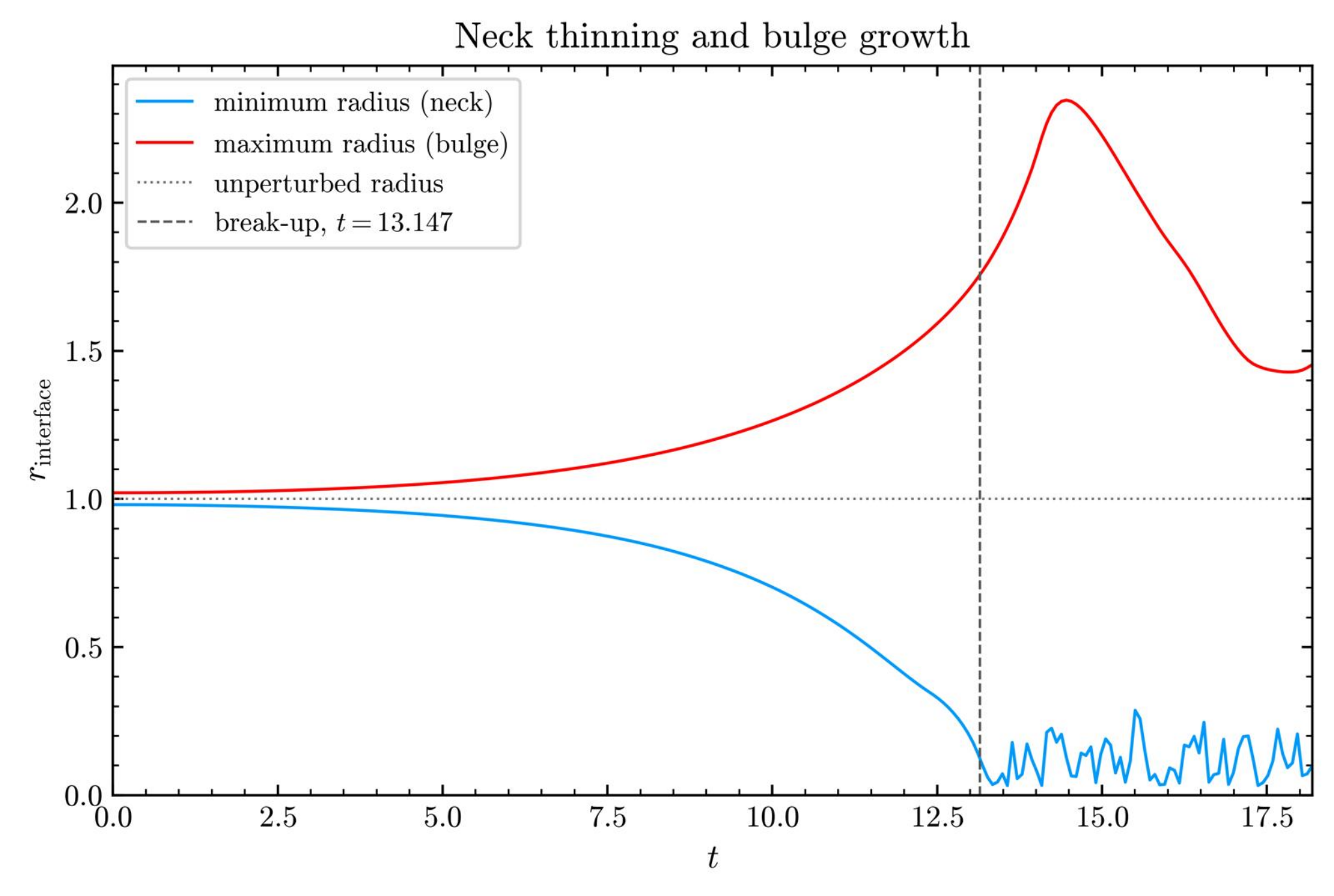

Interface growth over the 16-cell sweep (dotted: $a = 0.1R_0$)
$kR_0 = 0.2$
$kR_0 = 0.3$
$kR_0 = 0.4$
$kR_0 = 0.5$
$kR_0 = 0.6$
$kR_0 = 0.7$
$kR_0 = 0.8$
$kR_0 = 0.9$
$a(t)/a(0)$
10^1
10^0
0
5
10
15
20
25
30
t

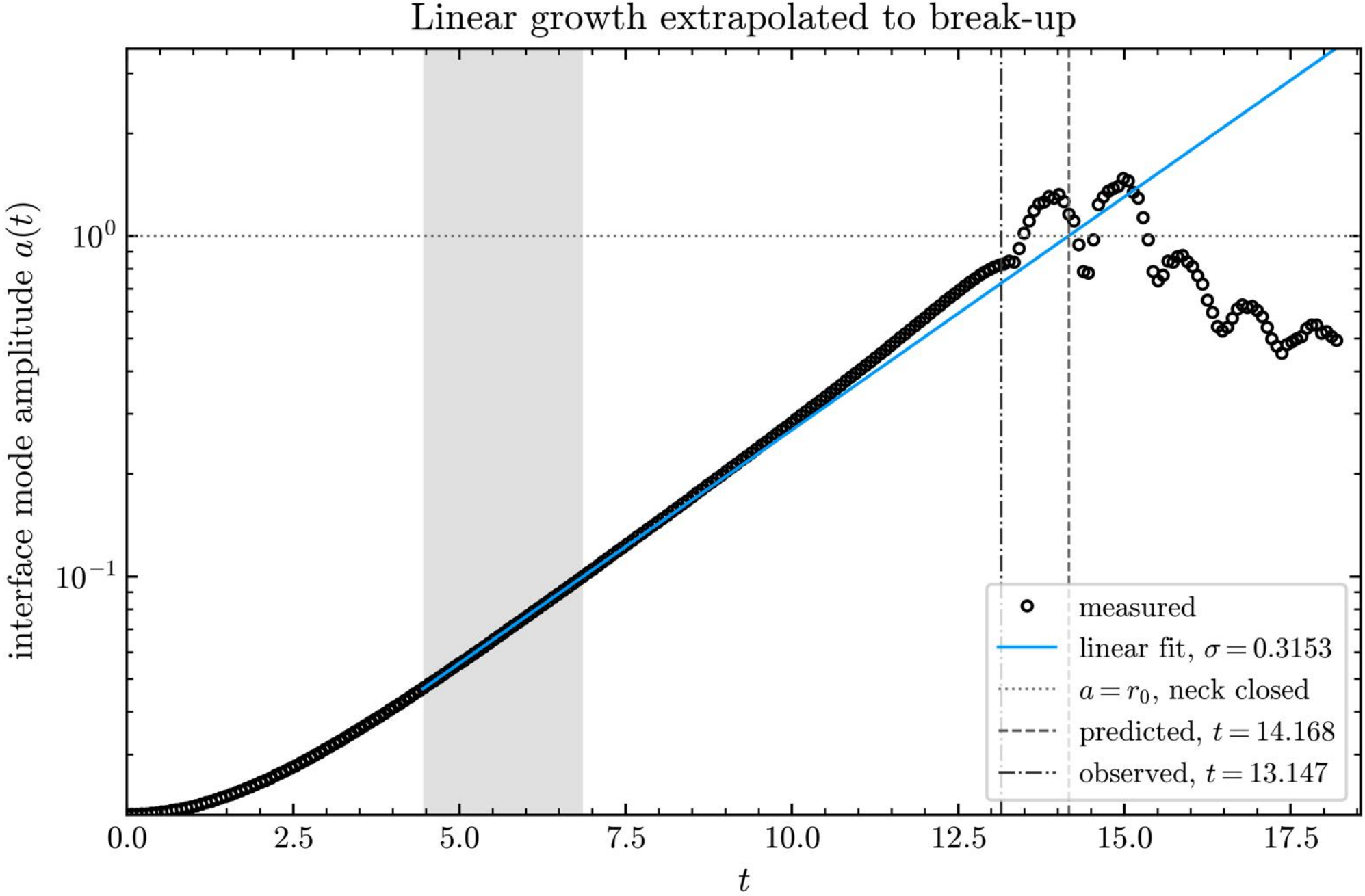

Linearity at $kR_0 = 0.7$, 16 cells per radius
$\epsilon = 0.0025$ (sw070‘16)
$\epsilon = 0.005$ (lin050‘16)
$\epsilon = 0.01$ (lin100‘16)
$a(t)/\epsilon$
10^1
10^0
departure from mean (%)
1
0
−1
0
2
4
6
8
10
12
t

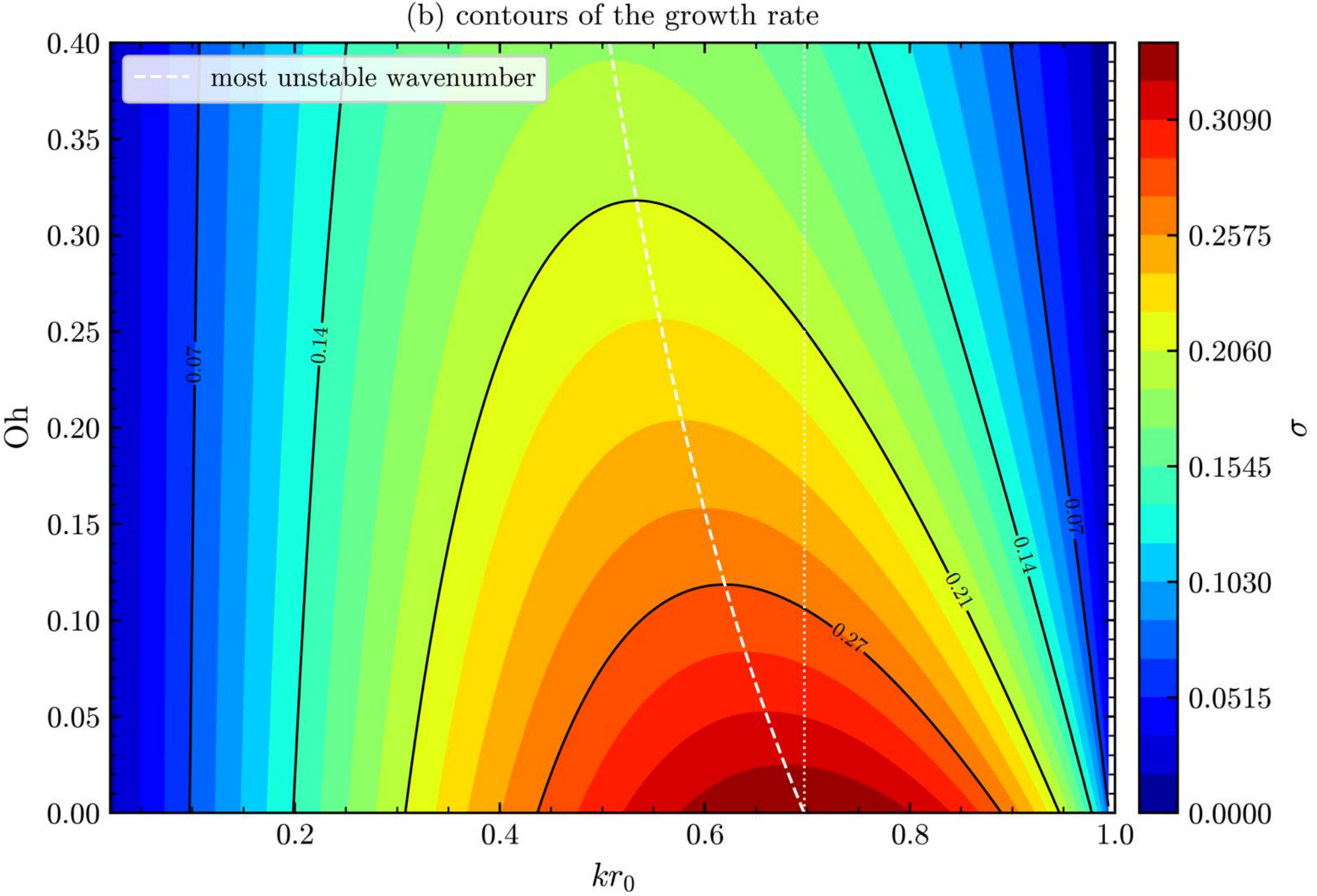

Axial velocity of the perturbation (twin difference) with the interface, meas16
$t = 0.07$
$\max|u_z'| = 1.1\text{e-}04$
$t = 3.59$
$\max|u_z'| = 7.4\text{e-}03$
$t = 7.11$
$\max|u_z'| = 2.5\text{e-}02$
$t = 10.63$
$\max|u_z'| = 8.3\text{e-}02$
z
r
$u_z'/\max|u_z'|$
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


1.0
0.5
0.0
−0.5
−1.0
$c(t)/r_0$ ($\times 100$)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
time t/t_c
measured, $kr_0 = 1.1$
fit: $\omega = 0.3283$, $\delta = 0.0329$
viscous theory: $\omega = 0.3295$, $\delta = 0.0334$

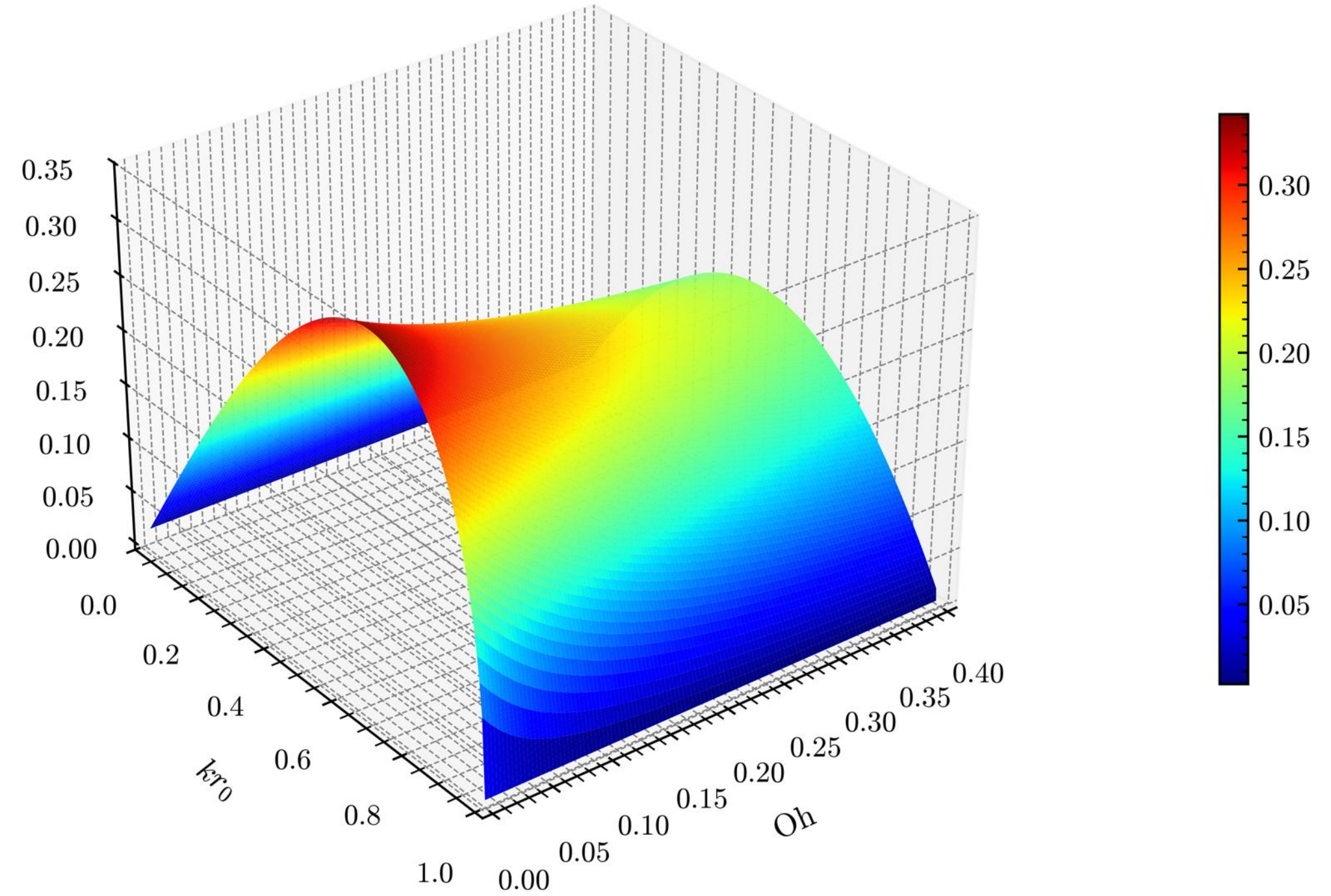

Interface through break-up
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$r_{\text{interface}}$
0
1
2
3
4
5
6
7
8
z
$t = 4.47$
$t = 6.19$
$t = 7.90$
$t = 9.69$
$t = 11.41$
$t = 13.12$

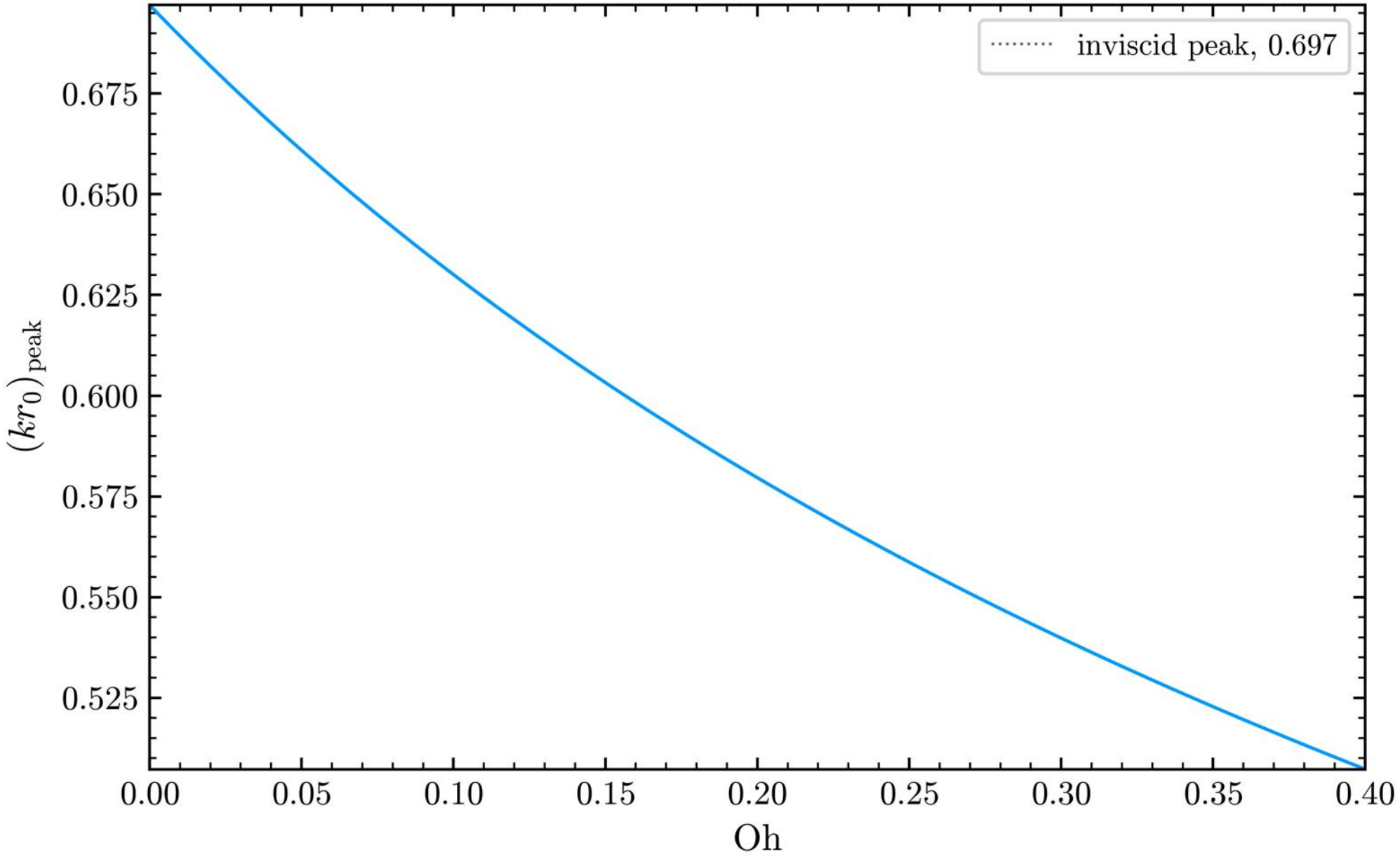

Neck thinning and bulge growth
minimum radius (neck)
maximum radius (bulge)
unperturbed radius
break-up, $t = 13.147$
$r_{\text{interface}}$
t
0.0
0.5
1.0
1.5
2.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5


Linear growth extrapolated to break-up
interface mode amplitude $a(t)$
10^0
10^{-1}
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
t
measured
linear fit, $\sigma = 0.3153$
$a = r_0$, neck closed
predicted, $t = 14.168$
observed, $t = 13.147$


(b) contours of the growth rate
most unstable wavenumber
0.07
0.14
0.21
0.27
0.14
0.07
Oh
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.2
0.4
0.6
0.8
1.0
kr_0
0.3090
0.2575
0.2060
0.1545
0.1030
0.0515
0.0000
σ

(a) growth rate over the parameter plane




inviscid peak, 0.697
0.675
0.650
0.625
0.600
0.575
0.550
0.525
$(kr_0)_{\text{peak}}$
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Oh